MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 6

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Recap of Previous Lecture

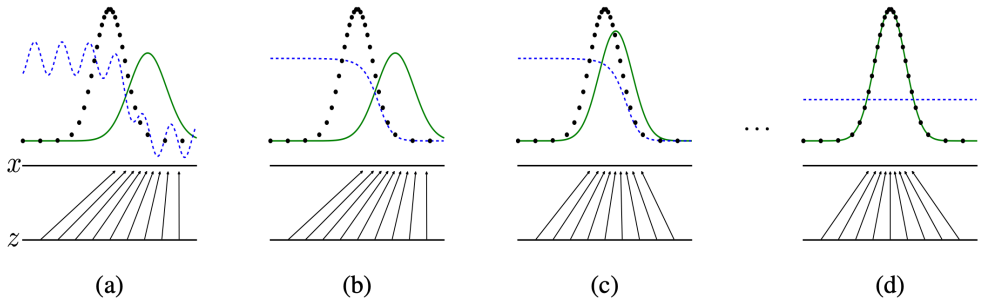
Objective

$$\min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi}} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\boldsymbol{\phi}}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z}))) \right]$$

Optimal Discriminator and Jensen-Shannon Divergence

For fixed G, write the bracketed objective as $V(G, D)$:

$$D^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\boldsymbol{\theta}}(\mathbf{x})}, \qquad V(G, D^*) = 2\, D_{\text{JS}}(p_{\text{data}}(\mathbf{x}) \,\|\, p_{\boldsymbol{\theta}}(\mathbf{x})) - 2 \log 2.$$



Mode collapse refers to the phenomenon where the generator in a GAN produces only one or a few different modes of the distribution.

Goodfellow I. J. et al. Generative Adversarial Networks, 2014
Metz L. et al. Unrolled Generative Adversarial Networks, 2016

Recap of Previous Lecture

GAN losses also train the reconstruction stage of a **latent generative model**.

Stage 1: Train the Autoencoder

A fixed pretrained network F defines the perceptual loss for reconstruction $\hat{\mathbf{x}}$:

$$\mathcal{L}_{\text{perc}} = \|F(\mathbf{x}) - F(\hat{\mathbf{x}})\|_2^2.$$

Train a discriminator D to separate real and reconstructed image patches; train the autoencoder to fool D.

A KL-autoencoder also uses weak KL regularization.

Stage 2: Freeze the Autoencoder and Learn a Prior

Fit the distribution of its latent codes — the **aggregated posterior**.

VQGAN: sample a code map $\mathbf{c}$ from an AR prior, look up its embeddings $\mathbf{z}_q(\mathbf{c})$, then decode.

$$\mathbf{c} \sim p_{\psi}(\mathbf{c}) \;\longrightarrow\; \mathbf{z}_q(\mathbf{c}) \;\longrightarrow\; \hat{\mathbf{x}}.$$

KL-autoencoder: a diffusion prior over continuous latents.

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \;\longrightarrow\; \mathbf{z} \sim p_{\psi}(\mathbf{z}) \;\longrightarrow\; \hat{\mathbf{x}}.$$

Esser P., Rombach R., Ommer B. Taming Transformers for High-Resolution Image Synthesis, 2021
Rombach R. et al. High-Resolution Image Synthesis with Latent Diffusion Models, 2022

Recap of Previous Lecture

Standard GAN Objective

$$\min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi}} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\boldsymbol{\phi}}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z}))) \right]$$

Theoretical Results

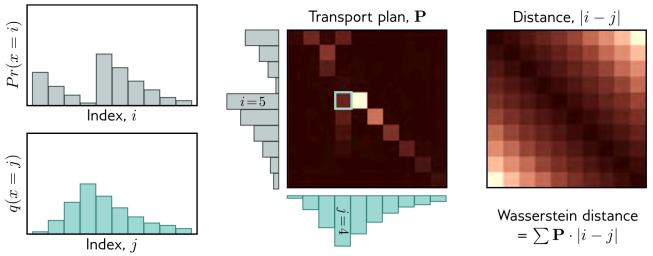
- Both the data distribution $p_{\text{data}}(\mathbf{x})$ and the generative distribution $p_{\boldsymbol{\theta}}(\mathbf{x})$ are supported on low-dimensional manifolds.
- If $p_{\text{data}}(\mathbf{x})$ and $p_{\boldsymbol{\theta}}(\mathbf{x})$ are disjoint, a smooth optimal discriminator can exist!

For such low-dimensional, disjoint manifolds:

$$D_{\text{KL}}(p_{\text{data}} \,\|\, p_{\boldsymbol{\theta}}) = D_{\text{KL}}(p_{\boldsymbol{\theta}} \,\|\, p_{\text{data}}) = \infty, \qquad D_{\text{JS}}(p_{\text{data}} \,\|\, p_{\boldsymbol{\theta}}) = \log 2$$

Weng L. From GAN to WGAN, 2019
Arjovsky M., Bottou L. Towards Principled Methods for Training Generative Adversarial Networks, 2017

Recap of Previous Lecture



Wasserstein Distance

$$W(\pi\|p) = \inf_{\gamma\in\Gamma(\pi,p)} \mathbb{E}_{(\mathbf{x}_1,\mathbf{x}_2)\sim\gamma}\|\mathbf{x}_1 - \mathbf{x}_2\|$$

$$= \inf_{\gamma\in\Gamma(\pi,p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\|\gamma(\mathbf{x}_1,\mathbf{x}_2)d\mathbf{x}_1 d\mathbf{x}_2$$

- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the transport plan: the amount of "dirt" assigned from $\mathbf{x}_1$ to $\mathbf{x}_2$.
- $\Gamma(\pi, p)$ denotes the set of all joint distributions $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ with marginals π and p ($\int \gamma(\mathbf{x}_1, \mathbf{x}_2)d\mathbf{x}_1 = p(\mathbf{x}_2)$, $\int \gamma(\mathbf{x}_1, \mathbf{x}_2)d\mathbf{x}_2 = \pi(\mathbf{x}_1)$).
- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the mass, $\|\mathbf{x}_1 - \mathbf{x}_2\|$ is the distance.

Simon J.D. Prince. Understanding Deep Learning, 2023
Arjovsky M., Chintala S., Bottou L. Wasserstein GAN, 2017

Recap of Previous Lecture

On a compact data space $\mathcal{X}$:

Dependence on Generator Parameters

For a feedforward generator $\mathbf{G}_{\boldsymbol{\theta}}$ with Lipschitz activations and $\mathbb{E}_{p(\mathbf{z})}\|\mathbf{z}\| < \infty$, $W(p_{\text{data}}\|p_{\boldsymbol{\theta}})$ is continuous in $\boldsymbol{\theta}$ and differentiable almost everywhere.

Convergence of Distributions

For a fixed distribution π and a sequence p_t on $\mathcal{X}$, as $t \to \infty$:

$$D_{\text{KL}}(\pi\|p_t) \to 0 \quad \Longrightarrow \quad D_{\text{JS}}(\pi\|p_t) \to 0 \quad \Longrightarrow \quad W(\pi\|p_t) \to 0$$

The first implication also holds for $D_{\text{KL}}(p_t\|\pi) \to 0$; the converses need not hold.

> **Wasserstein can capture distributions getting closer even when their supports do not overlap.**

Recap of Previous Lecture

Theorem (Kantorovich-Rubinstein Duality)

$$W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f(\mathbf{x}) \right],$$

where $f : \mathbb{R}^m \to \mathbb{R}$ is K-Lipschitz continuous ($\|f\|_L \leq K$).

WGAN Objective

$$\min_{\boldsymbol{\theta}} W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) \approx \min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi} \in \boldsymbol{\Phi}} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\boldsymbol{\phi}}(\mathbf{x}) - \mathbb{E}_{p(\mathbf{z})} f_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z})) \right].$$

- The discriminator D is replaced by function f: in WGAN, it is known as the **critic**, which is *not* a classifier.
- If the weights $\boldsymbol{\phi}$ are restricted to a compact set $\boldsymbol{\Phi} = [-c, c]^d$, then $f_{\boldsymbol{\phi}}(\mathbf{x})$ is K-Lipschitz continuous.

$$\begin{aligned} K \cdot W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) &= \max_{\|f\|_L \leq K} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f(\mathbf{x}) \right] \\ &\geq \max_{\boldsymbol{\phi} \in \boldsymbol{\Phi}} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\boldsymbol{\phi}}(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f_{\boldsymbol{\phi}}(\mathbf{x}) \right] \end{aligned}$$

Recap of Previous Lecture

Frechet Inception Distance (FID)

Use Inception image embeddings $\mathbf{z} = \mathbf{f}(\mathbf{x})$; compare Gaussian approximations of their distributions:

$$\begin{aligned}\mathrm{FID}(p_{\text{data}}, p_{\boldsymbol{\theta}}) = \|\boldsymbol{\mu}_{\text{data}} - \boldsymbol{\mu}_{\boldsymbol{\theta}}\|^2 \\ + \operatorname{tr}\left[\boldsymbol{\Sigma}_{\text{data}} + \boldsymbol{\Sigma}_{\boldsymbol{\theta}} - 2\left(\boldsymbol{\Sigma}_{\text{data}}^{1/2}\boldsymbol{\Sigma}_{\boldsymbol{\theta}}\boldsymbol{\Sigma}_{\text{data}}^{1/2}\right)^{1/2}\right].\end{aligned}$$

FID depends on the feature network, Gaussian approximation and sample size; it may not correlate with human evaluation.

CMMD: Comparing CLIP Image Embeddings

For CLIP image-embedding distributions P, Q, use a Gaussian kernel k:

$$\mathrm{MMD}_k^2(P, Q) = \mathbb{E}k(\mathbf{z}, \mathbf{z}') + \mathbb{E}k(\mathbf{w}, \mathbf{w}') - 2\mathbb{E}k(\mathbf{z}, \mathbf{w}).$$

Here $\mathbf{z}, \mathbf{z}' \sim P$ and $\mathbf{w}, \mathbf{w}' \sim Q$ are mutually independent. This compares more than means and covariances.

Heusel M. et al. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium, 2017
Jayasumana S. et al. Rethinking FID: Towards a Better Evaluation Metric for Image Generation, 2024

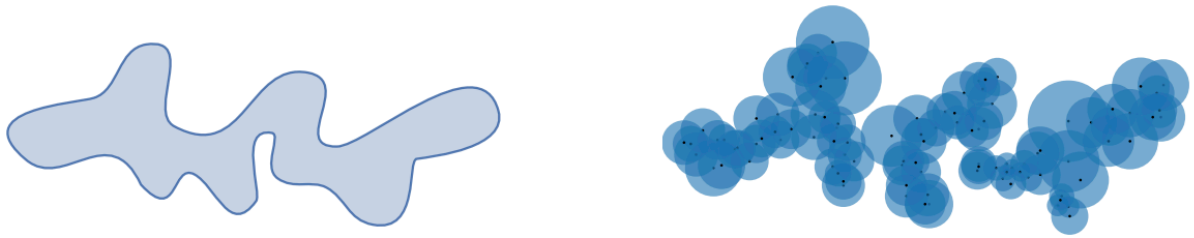
Recap of Previous Lecture

- $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$ - real samples;
- $\mathcal{S}_{\boldsymbol{\theta}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\boldsymbol{\theta}}(\mathbf{x})$ - generated samples.

Define a binary function:

$$\mathbb{I}(\mathbf{x}, \mathcal{S}) = \begin{cases} 1, & \text{if } \exists\, \mathbf{x}' \in \mathcal{S} : \|\mathbf{x} - \mathbf{x}'\|_2 \leq \|\mathbf{x}' - \text{NN}_k(\mathbf{x}', \mathcal{S})\|_2; \\ 0, & \text{otherwise.} \end{cases}$$

$$\text{Pr}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\boldsymbol{\theta}}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\boldsymbol{\theta}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\text{data}}); \quad \text{Rec}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\boldsymbol{\theta}}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\text{data}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\boldsymbol{\theta}}).$$



(a) True manifold (b) Approx. manifold

Embed the samples using a pretrained network (as in FID).

Recap of Previous Lecture

Score an image $\mathbf{x}$ for a text prompt t.

CLIP: Learn from Image–Caption Pairs

$$s_{\text{CLIP}}(\mathbf{x}, t) = \cos(\mathbf{f}_{\text{img}}(\mathbf{x}), \mathbf{f}_{\text{text}}(t)) \,.$$

PickScore: Learn from Human Choices

For the same prompt, a human prefers $\mathbf{x}^+$ to $\mathbf{x}^-$:

$$\mathbb{P}_{\boldsymbol{\theta}}(\mathbf{x}^+ \succ \mathbf{x}^- \mid t) = \frac{\exp r_{\boldsymbol{\theta}}(\mathbf{x}^+, t)}{\exp r_{\boldsymbol{\theta}}(\mathbf{x}^+, t) + \exp r_{\boldsymbol{\theta}}(\mathbf{x}^-, t)}.$$

Train the score $r_{\boldsymbol{\theta}}$ to assign higher probability to the human choice.

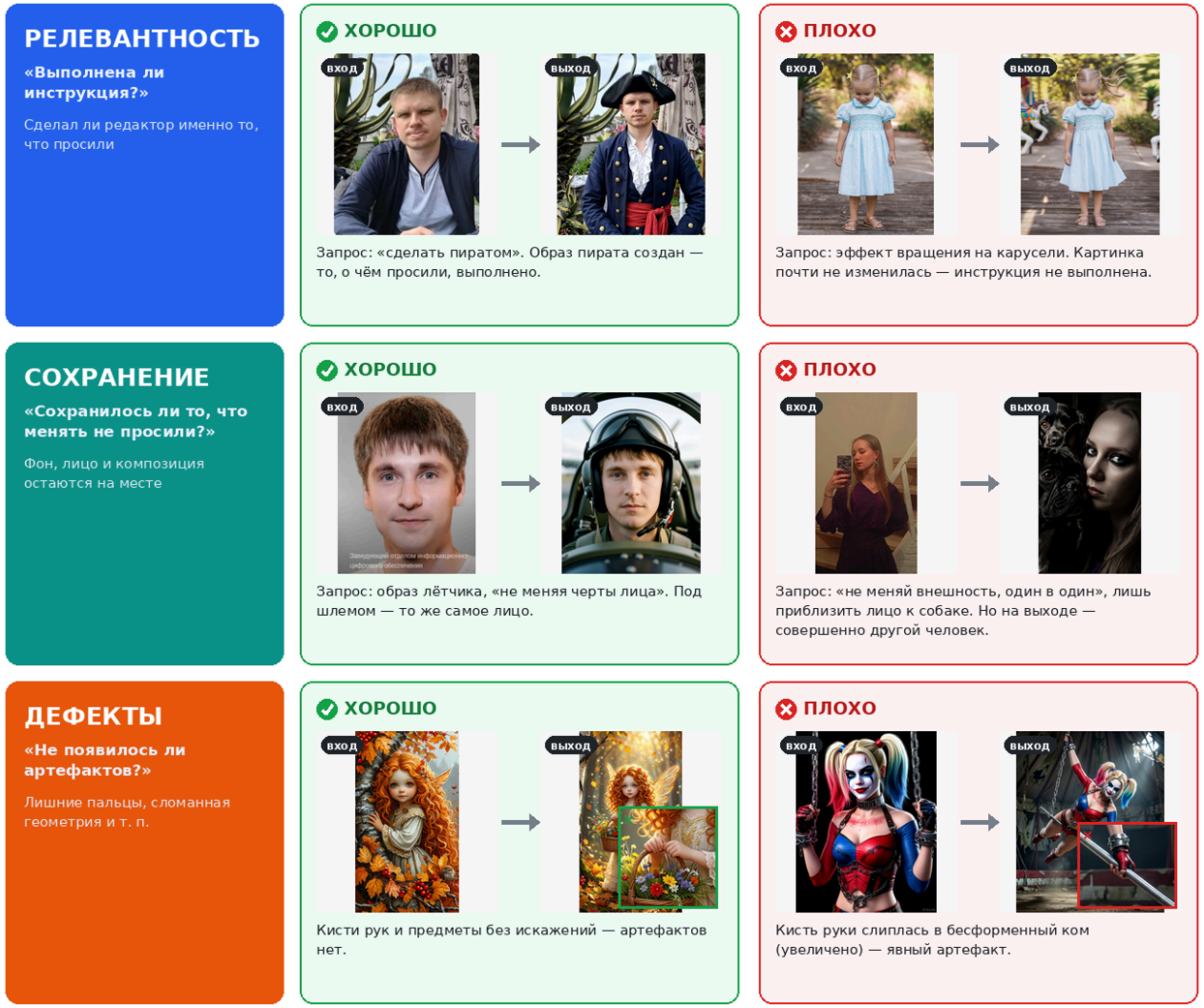
VQAScore / GenEval / DPG-Bench: evaluate compositional correctness; CLIP similarity can miss counts, attributes and relations.

A learned preference score can guide sample selection and model training — but optimizing the score can exploit its flaws.

Radford A. et al. Learning Transferable Visual Models From Natural Language Supervision, 2021
Kirstain Y. et al. Pick-a-Pic: An Open Dataset of User Preferences for Text-to-Image Generation, 2023
Lin Z. et al. Evaluating Text-to-Visual Generation with Image-to-Text Generation, 2024
Ghosh D. et al. GenEval, 2023 · Hu X. et al. ELLA (DPG-Bench), 2024

Recap of Previous Lecture

- No automated metric is perfect.
- The best way to evaluate generative models is by human assessment.
- It’s important to assess various properties.



Outline

01 Langevin Dynamics

02 Score Matching

Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

03 Forward Gaussian Diffusion Process

Outline

01 **Langevin Dynamics**

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Energy-Based Models

Unnormalized Density

$$p_{\boldsymbol{\theta}}(\mathbf{x}) = \frac{\hat{p}_{\boldsymbol{\theta}}(\mathbf{x})}{Z_{\boldsymbol{\theta}}}, \qquad \text{where } Z_{\boldsymbol{\theta}} = \int \hat{p}_{\boldsymbol{\theta}}(\mathbf{x}) d\mathbf{x}$$

- $\hat{p}_{\boldsymbol{\theta}}(\mathbf{x})$ can be any non-negative function.
- If we reparameterize as $\hat{p}_{\boldsymbol{\theta}}(\mathbf{x}) = \exp(-f_{\boldsymbol{\theta}}(\mathbf{x}))$, we eliminate the non-negativity constraint.

Log-Density Gradient

The gradient of the normalized log-density equals that of the unnormalized log-density:

$$\nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) = \nabla_{\mathbf{x}} \log \hat{p}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log Z_{\boldsymbol{\theta}} = \nabla_{\mathbf{x}} \log \hat{p}_{\boldsymbol{\theta}}(\mathbf{x})$$

- Suppose we already have this density (normalized or not) $p_{\boldsymbol{\theta}}(\mathbf{x})$.
- How can we sample from the model?

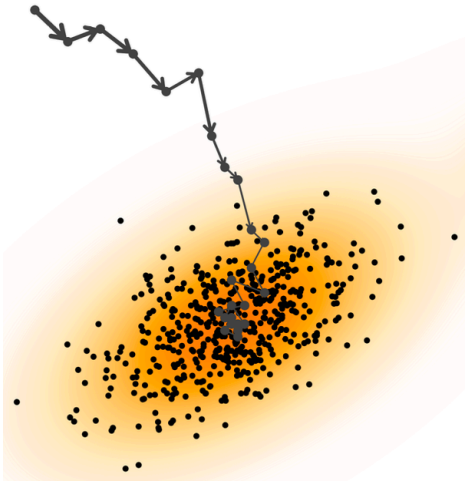
Langevin Dynamics

Theorem (Informal)

Let $\mathbf{x}_0$ be a random vector. Under mild regularity conditions, samples from the following dynamics will eventually follow $p_{\boldsymbol{\theta}}(\mathbf{x})$ (for sufficiently small η and large l):

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \qquad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}).$$

- What if $\boldsymbol{\epsilon}_l = \mathbf{0}$?
- The density $p_{\boldsymbol{\theta}}(\mathbf{x})$ is the **stationary** distribution of the Markov chain.
- The gradient is taken with respect to $\mathbf{x}$, not $\boldsymbol{\theta}$.
- $\nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x})$ defines a vector field.

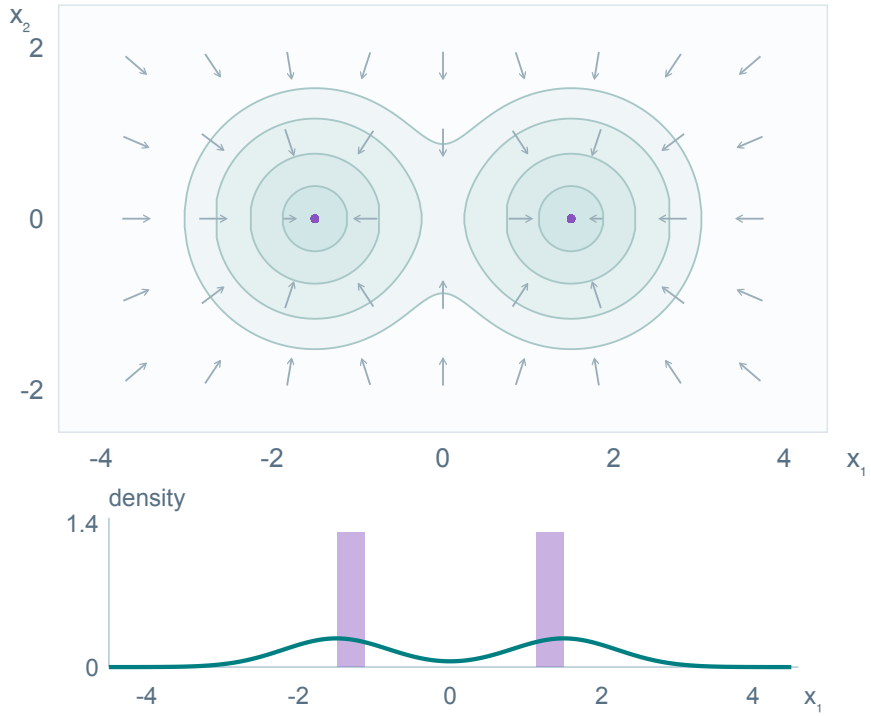


Lai C. H. et al. The principles of diffusion models, 2025.

Langevin Dynamics: Why Add Noise?

$$p_{\theta}(\mathbf{x}) = \tfrac{1}{2}\mathcal{N}(\boldsymbol{\mu}_{-}, 0.7^2\mathbf{I}) + \tfrac{1}{2}\mathcal{N}(\boldsymbol{\mu}_{+}, 0.7^2\mathbf{I}), \qquad \boldsymbol{\mu}_{\pm} = (\pm 1.5, 0), \quad \eta = 0.04.$$

Noise off: approach the modes Step 300



Density axis adapts to the histogram.

Noise on: explore the density Step 300



Density axis adapts to the histogram.

Finite step size and finite simulation time give approximate sampling.

Outline

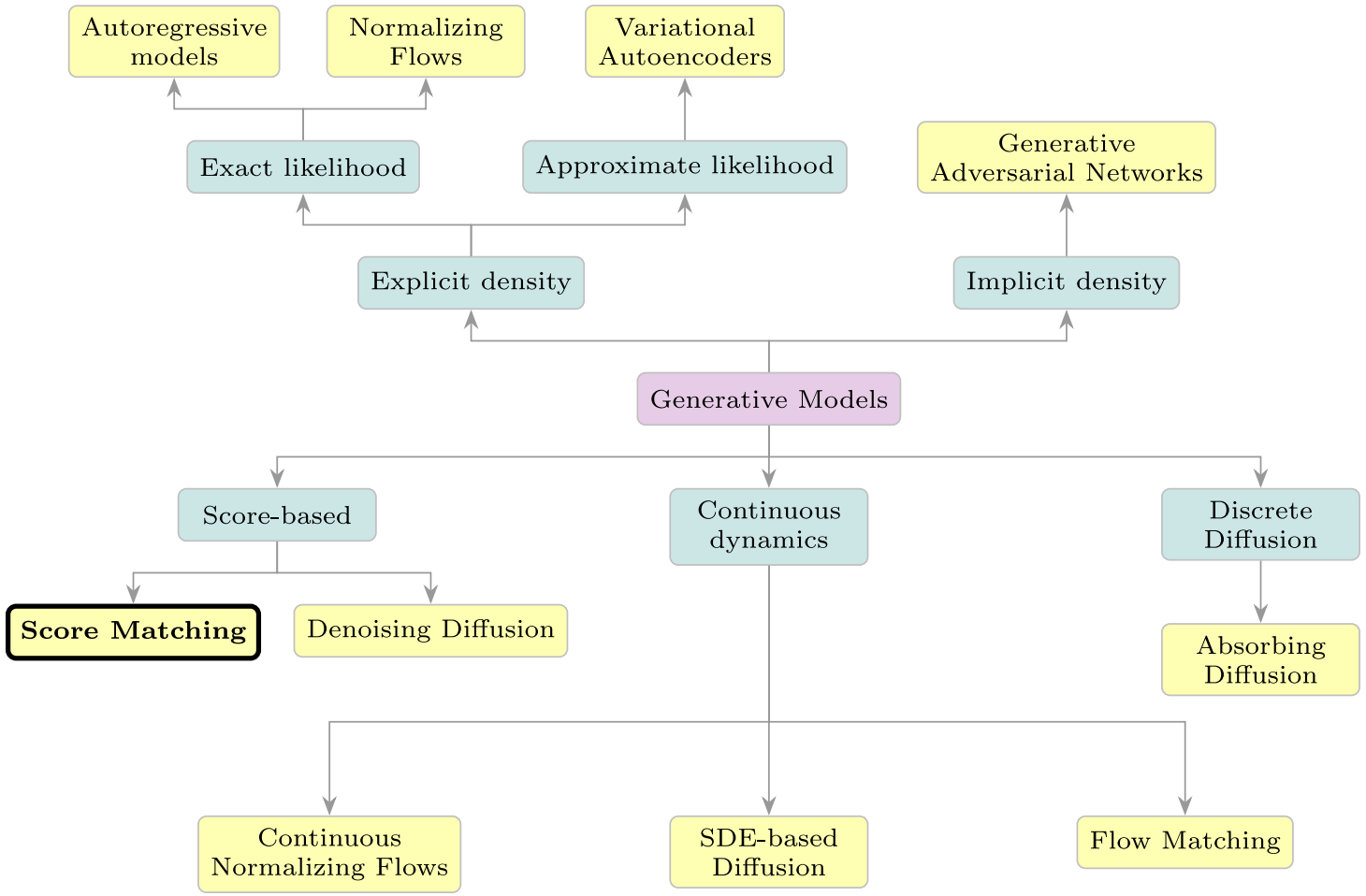
01 Langevin Dynamics

02 Score Matching

Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

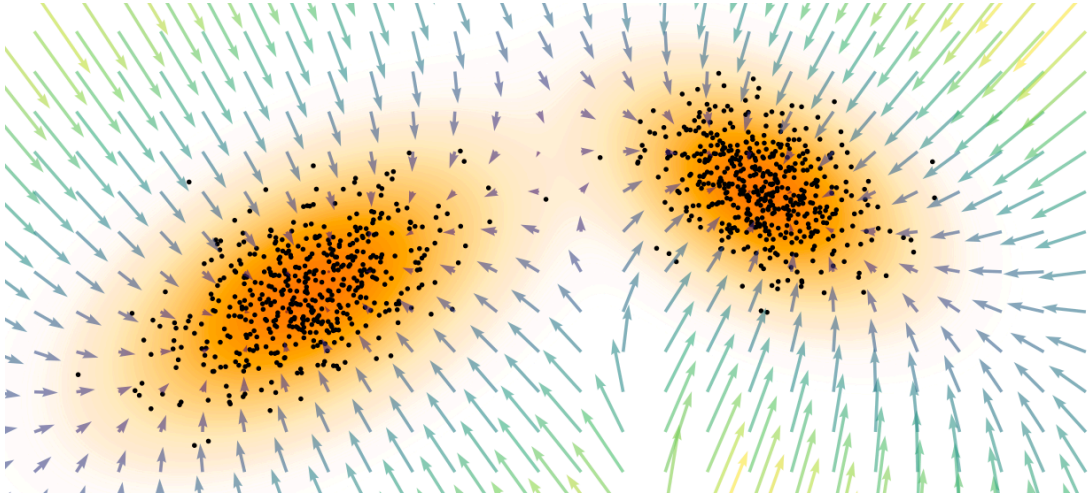
03 Forward Gaussian Diffusion Process

Generative Models Taxonomy



Score Function

$$\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) = \nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x})$$

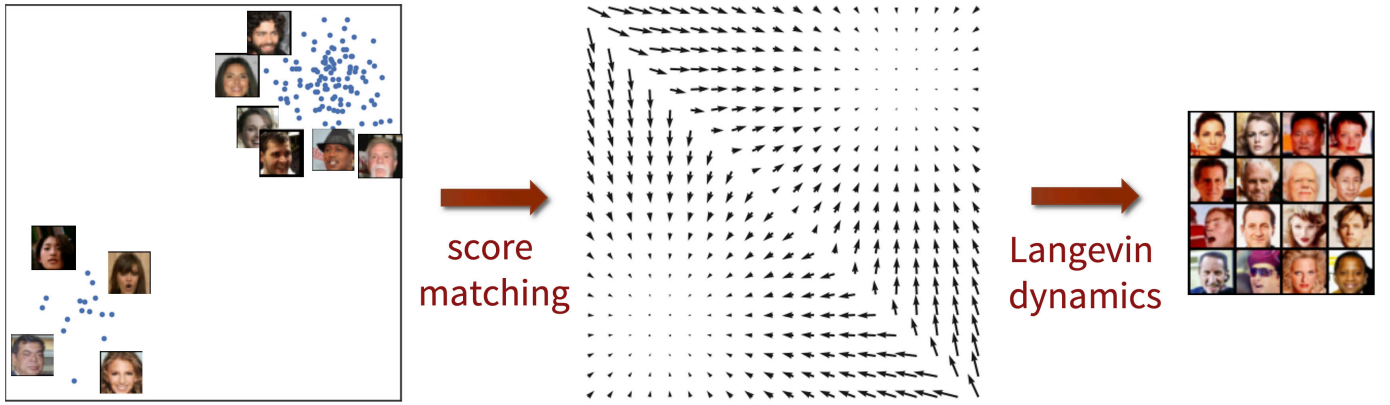


Langevin Dynamics

$$\begin{aligned}
\mathbf{x}_{l+1} &= \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\text{data}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l \\
&= \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l \\
&= \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l.
\end{aligned}$$

Lai C. H. et al. The principles of diffusion models, 2025.

Score Matching



$$\{\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N\} \overset{\text{i.i.d.}}{\sim} p(\mathbf{x}) \qquad \mathbf{s}_\theta(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p(\mathbf{x})$$

Fisher Divergence

$$\begin{aligned} D_F(p_{\text{data}}, p_{\boldsymbol{\theta}}) &= \frac{1}{2}\mathbb{E}_{p_{\text{data}}} \left\| \nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \\ &= \frac{1}{2}\mathbb{E}_{p_{\text{data}}} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}} \end{aligned}$$

Problem: We don't know $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$.

Why Is the Score Function Better than the Density?

Freedom from Normalization Constants

Many distributions are defined only up to an unnormalized density $\hat{p}(\mathbf{x})$:

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}) = \nabla_{\mathbf{x}} \log \hat{p}(\mathbf{x}) - \underbrace{\nabla_{\mathbf{x}} \log Z}_{=0} = \nabla_{\mathbf{x}} \log \hat{p}(\mathbf{x})$$

The score function bypasses the intractable constant Z entirely!

A Complete Representation

The score function fully characterizes the underlying distribution (up to a constant):

$$\log p(\mathbf{x}) = \log p(\mathbf{x}_0) + \int_0^1 \mathbf{s}(\mathbf{x}_0 + t(\mathbf{x} - \mathbf{x}_0))^\top (\mathbf{x} - \mathbf{x}_0) dt$$

Modeling the score is as expressive as modeling $p(\mathbf{x})$ itself, while often more tractable for generative modeling.

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Denoising Score Matching

Let us perturb the original data $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ with Gaussian noise:

$$\mathbf{x}_\sigma = \mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \quad q(\mathbf{x}_\sigma | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I})$$

$$q(\mathbf{x}_\sigma) = \int q(\mathbf{x}_\sigma | \mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x}.$$

Assumption

The solution to

$$\frac{1}{2} \mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta}, \sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

satisfies $\mathbf{s}_{\boldsymbol{\theta}, \sigma}(\mathbf{x}_\sigma) \approx \mathbf{s}_{\boldsymbol{\theta}, 0}(\mathbf{x}_0) = \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})$ if σ is sufficiently small.

- The score function of the noised data nearly matches the score function of the original data.
- The score function $\mathbf{s}_{\boldsymbol{\theta}, \sigma}(\mathbf{x}_\sigma)$ is parameterized by σ.
- **Note:** We don't know $q(\mathbf{x}_\sigma)$, just as we don't know $p_{\text{data}}(\mathbf{x})$.

Denoising Score Matching

Theorem

Under mild regularity conditions, the last term is independent of $\boldsymbol{\theta}$:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2$$
$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}$$

Gradient of the Noise Kernel

$$\mathbf{x}_\sigma = \mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}, \qquad q(\mathbf{x}_\sigma|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I})$$
$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) = -\frac{\mathbf{x}_\sigma - \mathbf{x}}{\sigma^2} = -\frac{\boldsymbol{\epsilon}}{\sigma}$$

- The right-hand side doesn't require computing $\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$ or even $\nabla_{\mathbf{x}_\sigma} \log p_{\text{data}}(\mathbf{x}_\sigma)$.
- $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)$ is trained to **denoise** the noised samples $\mathbf{x}_\sigma$.

Denoising Score Matching

Initial objective:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Noised objective:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}} \log q(\mathbf{x}_\sigma) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

This is equivalent to a denoising task:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathcal{N}(0,\mathbf{I})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}) + \frac{\boldsymbol{\epsilon}}{\sigma} \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \qquad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}).$$

Denoising Score Matching

Theorem

Under mild regularity conditions, the last term is independent of $\boldsymbol{\theta}$:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}.$$

Proof

For any integrable h, marginalize the joint distribution:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} h(\mathbf{x}_\sigma) = \int \left(\int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x} \right) h(\mathbf{x}_\sigma) d\mathbf{x}_\sigma = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} h(\mathbf{x}_\sigma).$$

Expand the squared error:

$$\begin{aligned} &\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 \\ = \mathbb{E}_{q(\mathbf{x}_\sigma)} &\left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) \right\|_2^2 + \underbrace{\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2}_{C_1} - 2\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right]. \end{aligned}$$

The score-norm term C_1 is independent of $\boldsymbol{\theta}$.

Denoising Score Matching

Theorem

Under mild regularity conditions, the last term is independent of $\boldsymbol{\theta}$:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}.$$

Proof (Continued)

Rewrite the mixed term:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right] = \int \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma) d\mathbf{x}_\sigma$$

$$= \int \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \left(\int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x} \right) d\mathbf{x}_\sigma = \iint p_{\text{data}}(\mathbf{x}) \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma|\mathbf{x}) \, d\mathbf{x} \, d\mathbf{x}_\sigma$$

$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right].$$

Substitute and complete the square:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left[\left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) \right\|_2^2 - 2 \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right] + C_1$$

$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}.$$

Denoising Score Matching

Denoising Score Matching Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

The optimal MSE predictor is the posterior mean of the regression target:

$$\mathbf{s}_{\boldsymbol{\theta}^*,\sigma}(\mathbf{x}_\sigma) = \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)} \left[\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right]$$

Marginal Score via Conditional Scores

$$\begin{aligned}
\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) &= \frac{\nabla_{\mathbf{x}_\sigma} \int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x}}{q(\mathbf{x}_\sigma)} \\
&= \int \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \cdot \frac{q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x})}{q(\mathbf{x}_\sigma)} d\mathbf{x} \\
&= \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)} \left[\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right]
\end{aligned}$$

Therefore: $\mathbf{s}_{\boldsymbol{\theta}^*,\sigma}(\mathbf{x}_\sigma) = \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$ - the optimal model learns the true **marginal** score!

Tweedie’s Formula

Let $\mathbf{x} \sim p_{\text{data}}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ be independent, with $\mathbf{x}_\sigma = \mathbf{x} + \sigma\boldsymbol{\epsilon}$ and $\sigma > 0$.

Using the posterior-mean identity from the previous slide:

$$\begin{aligned}\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) &= \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}\left[\frac{\mathbf{x} - \mathbf{x}_\sigma}{\sigma^2}\right] \\ &= \frac{\mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}[\mathbf{x}] - \mathbf{x}_\sigma}{\sigma^2}.\end{aligned}$$

Tweedie’s Formula

$$\mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}[\mathbf{x}] = \mathbf{x}_\sigma + \sigma^2 \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma).$$

Noise Prediction

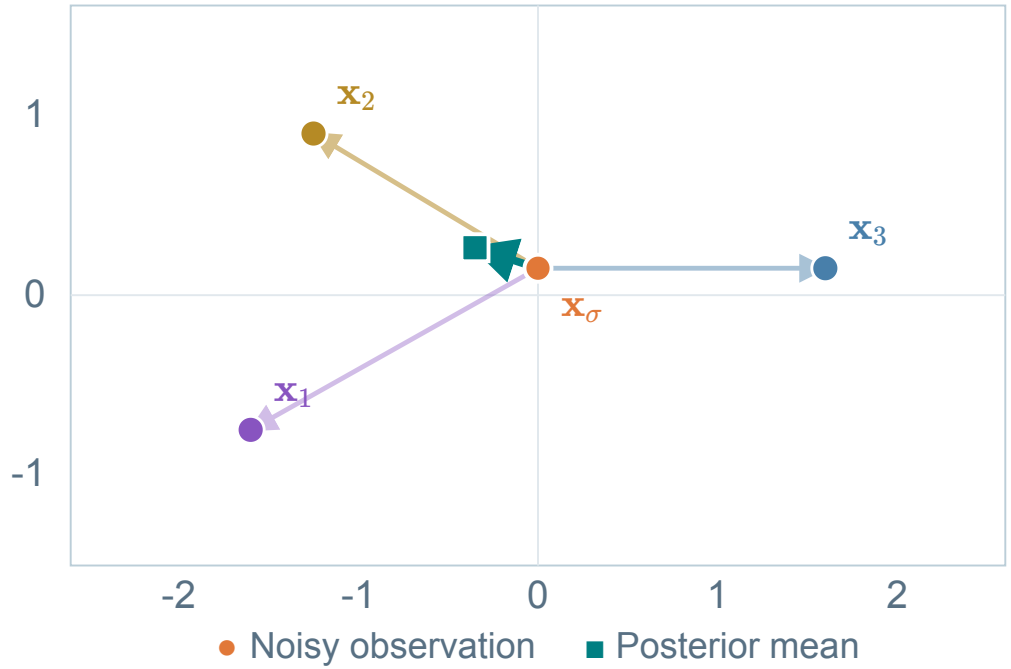
$$\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{x}_\sigma] = -\sigma \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma).$$

Both conditional means are MSE-optimal predictions given the noisy observation.

From Conditional Scores to Denoising

Toy data: three equally likely clean points $\mathbf{x}_i$, with $\mathbf{x}_\sigma = \mathbf{x} + \sigma\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ and $\sigma = 1$.

Fixed noisy observation between the clean-data modes.



Conditional scores

$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}_i) = (\mathbf{x}_i - \mathbf{x}_\sigma)/\sigma^2$$

Posterior weights

$q(\mathbf{x}_1\|\mathbf{x}_\sigma)$	$q(\mathbf{x}_2\|\mathbf{x}_\sigma)$	$q(\mathbf{x}_3\|\mathbf{x}_\sigma)$
22.9%	42.7%	34.4%

Marginal score = posterior-weighted average

$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) = \sum_i q(\mathbf{x}_i|\mathbf{x}_\sigma) \frac{\mathbf{x}_i - \mathbf{x}_\sigma}{\sigma^2}$$

Tweedie: from the score to the clean-data mean

$$\mathbb{E}[\mathbf{x}|\mathbf{x}_\sigma] = \mathbf{x}_\sigma + \sigma^2 \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$$

Posterior mean: (-0.35, 0.26)

> The MSE-optimal denoised prediction can lie between the clean-data modes.

Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010
Efron B. Tweedie's Formula and Selection Bias, 2011

Denoising Score Matching

Original objective:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Noisy objective:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}} \log q(\mathbf{x}_\sigma) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

This is equivalent to a denoising task:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathcal{N}(0,\mathbf{I})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x} + \sigma\boldsymbol{\epsilon}) + \frac{\boldsymbol{\epsilon}}{\sigma} \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \qquad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I})$$

Song Y. Generative Modeling by Estimating Gradients of the Data Distribution, blog post, 2021

Outline

01 Langevin Dynamics

02 Score Matching

Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

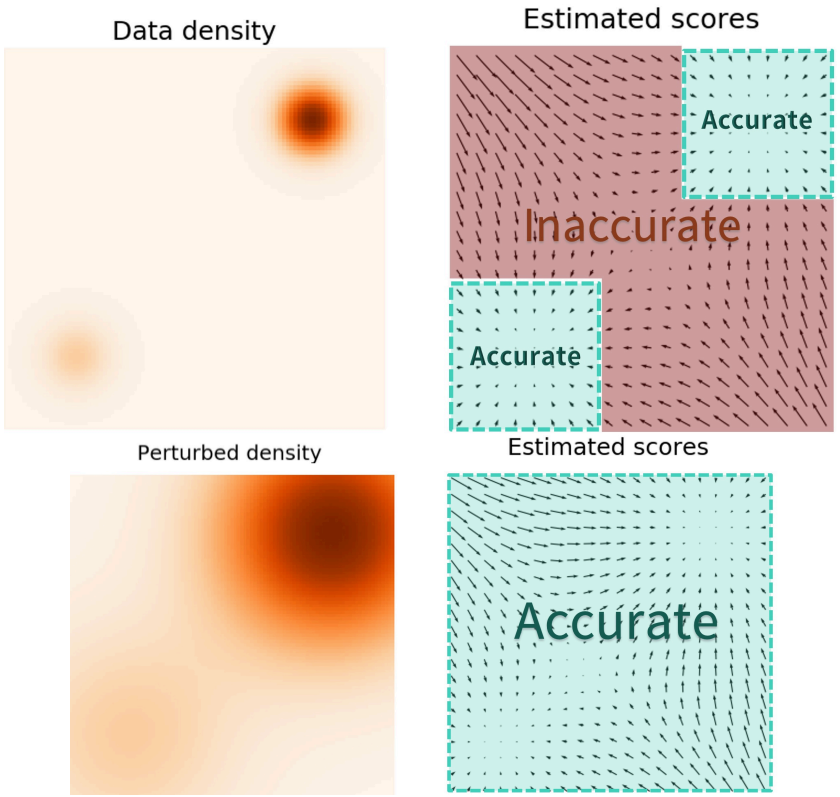
03 Forward Gaussian Diffusion Process

Denoising Score Matching

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathcal{N}(0,\mathbf{I})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}+\sigma\boldsymbol{\epsilon})+\frac{\boldsymbol{\epsilon}}{\sigma}\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2}\cdot\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta}\cdot\boldsymbol{\epsilon}_l$$

- For **small** σ, $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x})$ becomes inaccurate and Langevin dynamics fails to traverse modes.
- For **large** σ, robustness in low-density regions is achieved, but the model learns a distribution that is overly corrupted.

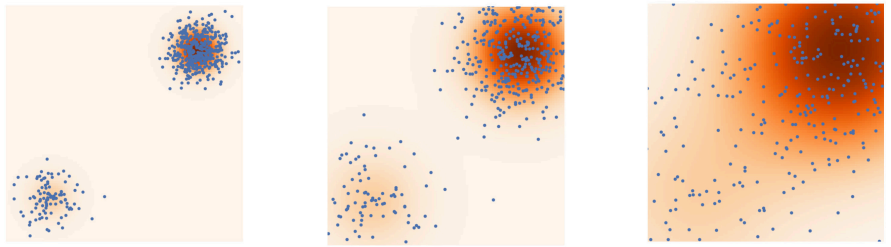


Noise-Conditioned Score Network (NCSN)

- Specify a sequence of noise levels: $\sigma_1 < \sigma_2 < \cdots < \sigma_T$.
- Perturb each data point with different noise levels: $\mathbf{x}_t = \mathbf{x} + \sigma_t \boldsymbol{\epsilon}$, so $\mathbf{x}_t \sim q(\mathbf{x}_t)$.
- Choose σ_1, σ_T such that:

$$q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \qquad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2 \mathbf{I})$$

$$\sigma_1 < \sigma_2 < \sigma_3$$





Noise-Conditioned Score Network (NCSN)

Train the denoising score function $\mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t)$ for each noise level σ_t using a unified weighted objective:

$$\sum_{t=1}^{T} \sigma_t^2 \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Here, $\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) = -\frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2} = -\frac{\boldsymbol{\epsilon}}{\sigma_t}$

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $t \sim U\{1, T\}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}$.
3. Compute loss $\mathcal{L} = \sigma_t^2 \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) + \frac{\boldsymbol{\epsilon}}{\sigma_t} \right\|^2$.
4. Update $\boldsymbol{\theta}$ via stochastic gradient descent.

How do we sample from such a model?

Noise-Conditioned Score Network (NCSN)

Sampling (Annealed Langevin Dynamics)

1. Sample $\mathbf{x}_T^0 \sim \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \approx q(\mathbf{x}_T)$.
2. Update $\mathbf{x}_t^l$ via L steps of Langevin dynamics at each noise level σ_t:
$$\mathbf{x}_t^l = \mathbf{x}_t^{l-1} + \frac{\eta_t}{2}\mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t^{l-1}) + \sqrt{\eta_t}\boldsymbol{\epsilon}_t^l.$$
3. Update $\mathbf{x}_{t-1}^0 := \mathbf{x}_t^L$ and proceed to the next noise level σ_{t-1}.



Outline

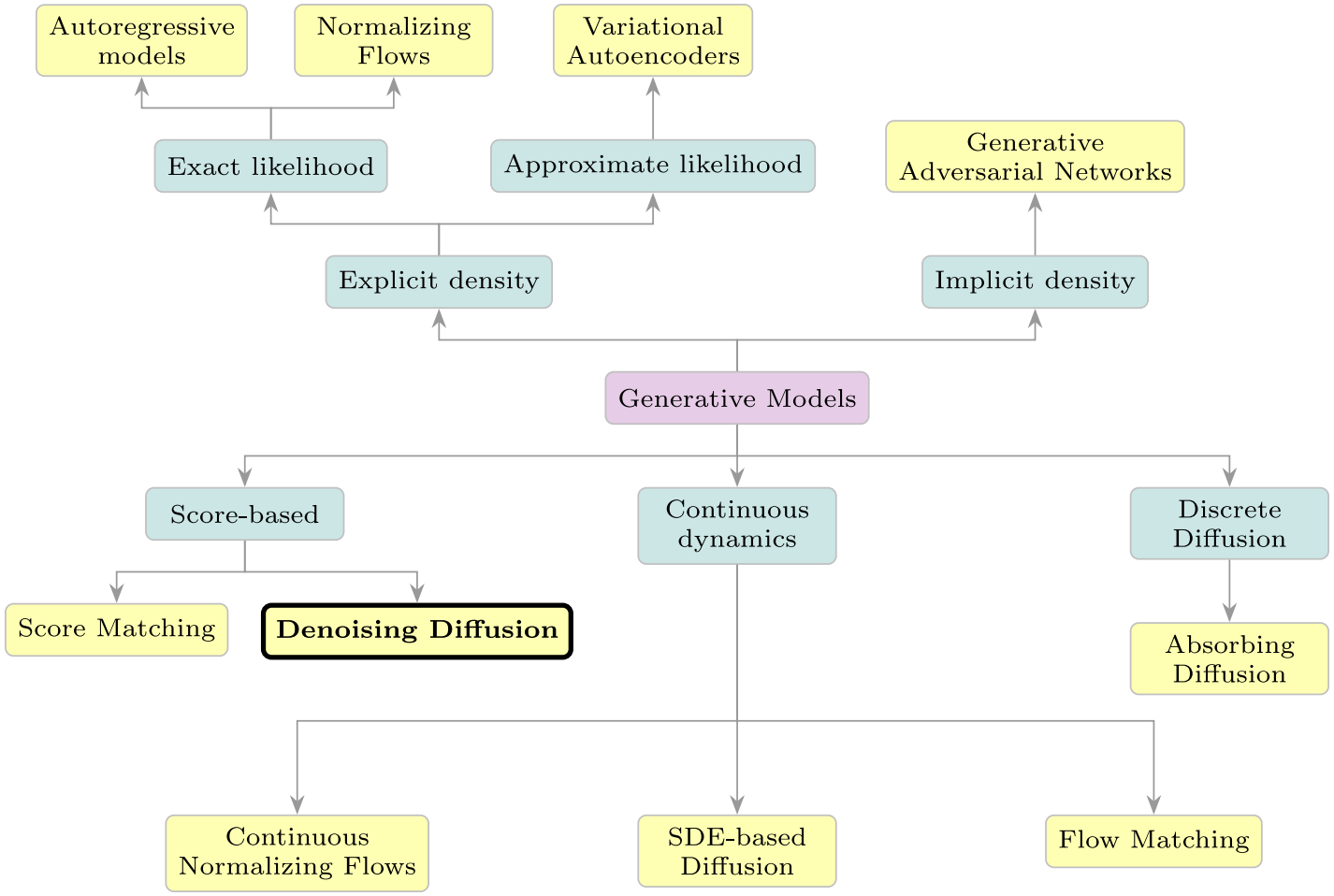
01 Langevin Dynamics

02 Score Matching

Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

03 Forward Gaussian Diffusion Process

Generative Models Taxonomy



Forward Gaussian Diffusion Process

Let $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$, $\beta_t \ll 1$. Define a Markov chain:

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta}\boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I})$$

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{1-\beta_t}\,\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t \\
&\approx \left(1 - \frac{\beta_t}{2}\right)\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t \\
&= \mathbf{x}_{t-1} + \frac{\beta_t}{2}(-\mathbf{x}_{t-1}) + \sqrt{\beta_t}\boldsymbol{\epsilon}_t
\end{aligned}$$

- $\beta_t = \eta$
- $\nabla_{\mathbf{x}_{t-1}} \log p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}) = -\mathbf{x}_{t-1} = \nabla_{\mathbf{x}_{t-1}} \log \mathcal{N}(0, \mathbf{I})$

Forward Gaussian Diffusion Process

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

Statement 1

Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s = \prod_{s=1}^{t}(1-\beta_s)$. Then

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})$$

Thus, samples at any timestep t can be generated directly from $\mathbf{x}_0$

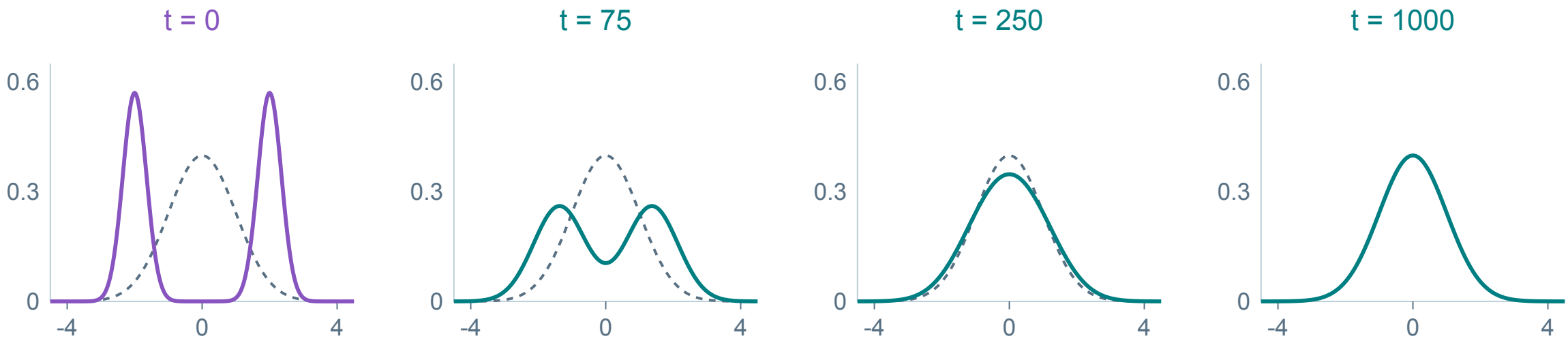
$$\begin{aligned}
\mathbf{x}_t &= \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t \\
&= \sqrt{\alpha_t}(\sqrt{\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_{t-1}}\boldsymbol{\epsilon}_{t-1}) + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + (\sqrt{\alpha_t(1-\alpha_{t-1})}\boldsymbol{\epsilon}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t) \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\boldsymbol{\epsilon}'_t \\
&= \ldots = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})
\end{aligned}$$

Forward Gaussian Diffusion Process

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right);$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}\right)$$

$$p_{\text{data}}(x_0) = \tfrac{1}{2}\mathcal{N}(-2, 0.35^2) + \tfrac{1}{2}\mathcal{N}(2, 0.35^2), \qquad \beta_t = 0.01$$



---- Standard Gaussian reference $\mathcal{N}(0,1)$.

The plots show marginal densities; the kernels above are conditional.

Forward Gaussian Diffusion Process

Statement 2

Applying the Markov chain to any distribution $p_{\text{data}}(\mathbf{x})$ yields $\mathbf{x}_\infty \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$, the **stationary** (limiting) distribution:

$$p_\infty(\mathbf{x}) = \int q(\mathbf{x}|\mathbf{x}')p_\infty(\mathbf{x}')d\mathbf{x}'$$

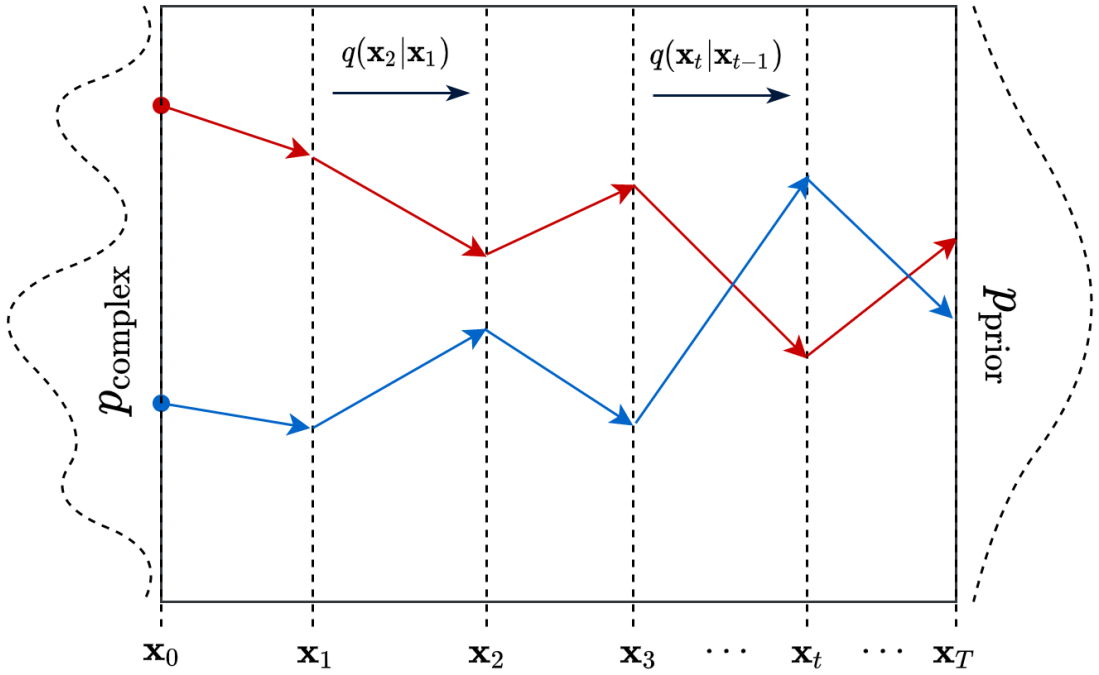
$$p_\infty(\mathbf{x}) = \int q(\mathbf{x}_\infty|\mathbf{x}_0)p_{\text{data}}(\mathbf{x}_0)d\mathbf{x}_0$$

$$\approx \mathcal{N}(0, \mathbf{I}) \int p_{\text{data}}(\mathbf{x}_0)d\mathbf{x}_0 = \mathcal{N}(0, \mathbf{I})$$

Note: This holds iff $\bar{\alpha}_t \to 0$, i.e., $\sum_{t=1}^{\infty} \beta_t = +\infty$.

Forward Gaussian Diffusion Process

Diffusion describes the migration of particles from regions of high density to those of low density.



1. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$
2. $\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t,$
   $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I}),\ t \geq 1$
3. After $T \gg 1$ steps: $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$

If this process can be reversed, we can sample from $p_{\text{data}}(\mathbf{x})$ by starting from noise $p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$.
Our goal now becomes inverting this diffusion.

Denoising Score Matching

NCSN

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma_t^2\mathbf{I}), \quad q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2\mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) = -\frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2}$$

Gaussian Diffusion

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}), \quad q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) = -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{1-\bar{\alpha}_t}$$

Theorem (Denoising Score Matching)

$$\mathbb{E}_{q(\mathbf{x}_t)} \left\| \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t) \right\|_2^2$$
$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) \right\|_2^2 + \text{const}$$

Note: Annealed Langevin dynamics applies to diffusion, too.
NCSN with annealed Langevin dynamics is an early score-based generative framework (2019–2020).

Summary

- Langevin dynamics combines log-density gradients and Gaussian noise to sample approximately from an unnormalized model.
- Score matching minimizes Fisher divergence to estimate the score function.
- Denoising score matching makes the score objective estimable using corrupted samples and the known corruption kernel.
- Tweedie’s formula converts the score into MSE-optimal clean-data and noise predictions.
- Noise-Conditioned Score Networks learn scores at multiple noise levels and sample with annealed Langevin dynamics.
- The forward Gaussian diffusion process admits a closed-form conditional distribution for any timestep.
- With a suitable noise schedule, the forward process converges to standard Gaussian noise; denoising score matching also applies to these diffusion marginals.